

Izmir Institute of Technology

Faculty of Engineering
Computer Engineering

THESIS / PROJECT PROPOSAL

AI-Driven Aging-Aware Adaptive Time Quantum Scheduling

Reliable and Efficient Scheduler Design with Cross-Layer Reliability
Feedback

Student Name:	Omer Faruk Teker
Student ID:	To be specified
Supervisor:	To be specified
Program:	BSc
Department:	Computer Engineering
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Abstract

This research proposal aims to unify two lines of work that are usually treated separately: *adaptive time quantum selection* in CPU scheduling and *aging-aware runtime control* in hardware reliability management. Existing Round Robin variants mainly optimize waiting time, turnaround time, and context-switch overhead through workload-aware quantum adaptation, while reliability-aware systems focus on thermal management, DVFS, task remapping, and online monitoring to preserve hardware lifetime [1, 2, 3, 6, 9]. However, only limited work directly incorporates aging and thermal feedback into scheduler-level time quantum control.

The proposed research introduces an *aging-aware adaptive scheduler* that combines queue-state, thermal telemetry, and aging estimates within a single control loop. The methodology is organized in three stages: (i) heuristic aging-penalized quantum selection, (ii) physics-informed supervised learning for quantum prediction, and (iii) contextual bandit or reinforcement learning for long-horizon policy optimization. The core hypothesis is that such a scheduler can achieve either lower aging accumulation at comparable performance or higher performance under the same aging budget.

Keywords: adaptive scheduling, Round Robin, time quantum, hardware aging, reliability-aware control, reinforcement learning

1. Introduction

Time quantum selection is one of the key design decisions in time-shared schedulers. When the quantum is too small, context-switch overhead increases and throughput may drop. When it is too large, response time and interactivity suffer. For this reason, a large body of dynamic Round Robin literature has proposed workload-aware quantum selection strategies based on mean, median, quartile, or percentile statistics [1, 2, 3, 4].

At the same time, scheduler decisions indirectly affect hardware lifetime. Imbalanced core usage, persistent hotspots, long periods of high utilization, and strong thermal gradients can accelerate degradation mechanisms such as NBTI, PBTI, HCI, electromigration, and thermal cycling [6, 7, 9]. Despite this connection, scheduling literature and reliability-aware runtime control literature have largely evolved independently.

This proposal addresses that gap by asking whether a scheduler can select time quantum values using not only queue-state information but also thermal and aging feedback.

2. Problem Statement

The central problem of this research can be stated as follows:

Fixed or purely queue-statistics-driven dynamic time quantum selection does not explicitly account for the long-term hardware aging and reliability cost of scheduling decisions. Therefore, a new scheduler-level control mechanism is needed to jointly optimize performance, fairness, thermal behavior, and projected lifetime.

This problem is particularly important in multicore systems, where task placement, context-switch rhythm, and core utilization asymmetry have temporal and spatial effects on degradation behavior.

3. Research Questions and Hypotheses

3.1. Research Questions

1. Can time quantum selection achieve a better performance-reliability trade-off when queue-state and thermal/aging telemetry are used together?
2. Can an aging-aware adaptive scheduler produce lower aging scores than fixed RR and classical dynamic RR under similar throughput levels?
3. Can physics-informed supervised learning or RL-based policies generalize better than heuristic policies under workload variation?

4. Does extending the scheduler action space with migration and DVFS lead to measurable projected lifetime improvements?

3.2. Hypotheses

1. Queue-aware and aging-aware hybrid schedulers provide a better latency-reliability trade-off than traditional dynamic RR schedulers.
2. Incorporating thermal and aging state into quantum selection reduces hotspot behavior and cumulative stress imbalance.
3. Physics-informed supervised models can select better time quanta than purely statistical schedulers.
4. RL-based control can reduce long-term aging cost when combined with suitable rewards and safety constraints.

4. Significance and Expected Contribution

This research is significant for four reasons:

1. It bridges scheduling literature and aging-aware reliability literature.
2. It treats time quantum selection as both a performance problem and a hardware health problem.
3. It compares heuristic, supervised learning, and RL-based methods in a shared simulation environment.
4. It directly tests the claims of “same performance, less aging” and “same aging budget, better performance.”

5. Literature Review

5.1. Adaptive Time Quantum Scheduling

Dynamic quantum selection has long been studied in the context of Round Robin scheduling. Behera et al. reported improvements with a re-adjusted dynamic quantum policy [1]. Noon et al. proposed a mean-based quantum computation strategy [2], while more recent work has explored median-average and percentile-based approaches [3, 4]. Although these studies demonstrate the value of workload-aware quantum selection, they generally omit thermal and aging effects.

5.2. AI-Driven Scheduling

Machine learning is increasingly used in scheduling research, but direct AI-driven CPU quantum selection remains relatively sparse. Chitaliya et al. framed quantum optimization as a prediction problem and evaluated ML-based quantum estimation [5]. Still, few works directly address scheduler-level reliability-aware quantum control.

5.3. Hardware Aging and Reliability-Aware Control

Srinivasan et al. established the importance of lifetime reliability-aware microprocessor design [6]. Agarwal et al. highlighted the role of failure prediction in transistor aging management [7], while Kumar et al. developed adaptive techniques to compensate for aging-induced performance degradation [9]. Reliability-aware energy management and RL-based DVFS demonstrate that runtime performance-lifetime trade-offs can indeed be controlled at higher system layers [8, 10].

5.4. Research Gap

The key gaps in the literature are the following:

- Quantum selection studies do not typically use hardware aging information.
- Aging-aware runtime control studies rarely optimize scheduler quantum directly.
- AI-based scheduler research rarely adopts a reliability-centric objective.

This proposal is positioned precisely at this intersection.

6. Proposed Methodology

6.1. General Framework

The proposed system consists of four components:

1. Workload generator or trace source
2. Scheduler engine
3. Thermal estimator and aging estimator
4. Logging, evaluation, and analysis layer

The policy layer will use queue-state, thermal telemetry, and aging estimates to generate quantum, migration, and optionally DVFS decisions.

6.2. Method 1: Heuristic Aging-Penalized Quantum

The first method is an interpretable baseline:

$$Q_t = f(S_t^{queue}) - \lambda_1 g(S_t^{thermal}) - \lambda_2 h(S_t^{aging})$$

Here, f produces a base quantum using mean, median, or quartile statistics, while g and h provide penalty terms derived from temperature, hotspot severity, thermal gradient, cumulative stress, and estimated slack loss.

6.3. Method 2: Physics-Informed Supervised Learning

In the second stage, optimal time quantum selection is formulated as a prediction problem. For each workload window, candidate quantum values are evaluated through a combined objective:

$$J = w_1 \cdot \text{Latency} + w_2 \cdot \text{ContextSwitchCost} + w_3 \cdot \text{ThermalCost} + w_4 \cdot \text{AgingCost}$$

The quantum that minimizes J becomes the training label. Models such as Random Forest, XGBoost, and LightGBM will then be used to predict quantum values. The approach is called physics-informed because both labeling and feature design explicitly encode thermal and aging meaning.

6.4. Method 3: Contextual Bandit and Reinforcement Learning

In the third stage, scheduling is treated as a sequential decision problem. The state space includes queue length, remaining burst statistics, per-core utilization, current temperature, temperature gradient, cumulative aging score, previous quantum values, and migration history. A representative reward function is:

$$R_t = \alpha \cdot \text{Throughput}_t - \beta \cdot \text{ResponseTime}_t - \gamma \cdot \text{ContextSwitches}_t - \delta \cdot \text{HotspotPenalty}_t - \eta \cdot \text{AgingIncrement}_t$$

Algorithms such as DQN, Double DQN, or PPO may be used. All learned policies will be deployed under a safety filter that enforces peak-temperature thresholds, starvation prevention, and migration-rate limits.

7. Experimental Design

7.1. Simulation Environment

The initial research platform will be a Python-based discrete-event scheduler simulator. This choice supports rapid experimentation, reproducibility, and fair comparison of policy variants. Integration with gem5 or Sniper may be considered in later phases.

7.2. Workload Classes

Table 1: Planned workload classes.

Class	Type	Description
Short CPU-bound	Synthetic	Short bursts; high response-time sensitivity
Long CPU-bound	Synthetic	Long bursts; strong sensitivity to quantum size
Heavy-tail bursts	Synthetic	Pareto-like burst distribution
Bursty arrivals	Synthetic	Clustered arrivals that create queue pressure
Periodic tasks	Synthetic	Real-time-like periodic workloads
Hot-core stress	Scenario	Workload shaped to accumulate stress on one core
Balanced multi-core mix	Scenario	Suitable for observing wear-leveling behavior
Trace-driven mix	Realistic	Derived from scheduler traces if available

7.3. Baselines

1. Fixed Round Robin
2. Mean-based dynamic RR
3. Median/quartile-based dynamic RR
4. Thermal-aware scheduler
5. Aging-aware but non-AI scheduler
6. Proposed heuristic hybrid scheduler
7. Proposed supervised scheduler
8. Proposed bandit / RL scheduler

7.4. Thermal and Aging Models

The thermal model is defined in a simple but consistent form:

$$T_{t+1} = aT_t + bU_t + cS_t + dT_{amb}$$

where U_t is utilization, S_t is a switching proxy, and T_{amb} is ambient temperature. The aging increment is modeled as:

$$\Delta A_t = k_1 e^{\rho T_t} + k_2 U_t^p + k_3 C_t$$

where C_t captures additional stress related to context-switch or switching-rate behavior. In later phases, BTI-like, HCI-like, EM-proxy, and thermal-cycling components may be tracked separately.

8. Evaluation Metrics

8.1. Scheduler Performance

- Average waiting time
- Average turnaround time
- Average response time
- Throughput
- Context-switch count
- Fairness and starvation indicators

8.2. Reliability and Thermal Metrics

- Peak temperature
- Mean temperature per core
- Thermal gradient
- Thermal-cycling severity
- Cumulative aging score
- Projected lifetime proxy

8.3. Primary Comparative Claims

The proposal directly tests two core claims:

- lower aging accumulation at comparable performance

- better performance under the same aging budget

9. Illustrative Scenario and Objective Expectation Framework

This section is added to make the proposal more concrete under a realistic operating scenario and to present both the expected benefits and the possible limitations in an objective way.

9.1. Illustrative Scenario

Consider a four-core system with a mixed ready queue containing both short and long CPU-bound jobs. Assume that, within a given scheduling window, the following conditions hold:

- Core 1 has been under high utilization for a long period.
- Core 1 temperature is higher than the others.
- The aging score of Core 1 is increasing faster than the rest.
- Queue occupancy is high, meaning that the scheduler is under performance pressure.

In this situation, a conventional dynamic RR scheduler would likely increase the quantum in order to reduce queue pressure, but because it does not incorporate hardware health information, it may further intensify the stress on the same core. By contrast, the proposed aging-aware scheduler would evaluate one or more of the following actions jointly:

1. increase the quantum according to queue pressure, but limit that increase via thermal/aging penalties
2. migrate jobs to a less stressed core
3. use DVFS to move the stressed core into a more conservative operating mode if needed
4. filter migration or quantum decisions if fairness constraints are threatened

This scenario makes clear that the proposed system is intended to answer not only “which job should run and for how long,” but also “what is the hardware-health consequence of this scheduling decision?”

9.2. Potential Positive Outcomes

Under such a scenario, the following positive outcomes may be expected:

- reduction in hotspot behavior on specific cores
- more balanced core-wise aging score distribution

- lower aging accumulation at comparable throughput relative to fixed RR or queue-only dynamic RR
- lower thermal gradients and peak temperatures
- improved projected lifetime proxy in long-running experiments

9.3. Potential Negative Outcomes and Limitations

For an objective evaluation, the proposal must also acknowledge the possibility of limited or negative outcomes:

- migration and policy-computation overhead may increase total latency
- overly aggressive aging protection may hurt throughput or turnaround time
- thermal telemetry alone may not fully represent aging behavior, making the proxy model incomplete
- RL-based policies may be sensitive to training data and reward weights
- benefits observed in simulation may shrink in real-system deployment
- for some workload families, classical dynamic RR may already perform well enough that additional gains are small

9.4. Objective Success Criteria

The work should be considered successful only if it can demonstrate outcomes such as:

1. statistically meaningful aging-score reduction under comparable throughput
2. lower waiting time or turnaround time under the same aging budget
3. clear reduction in peak temperature or hotspot count
4. consistent improvement across multiple workload classes

If these criteria are not fully satisfied, the study may still make a valuable contribution by clearly identifying the conditions under which scheduler-level aging-aware control is beneficial and the conditions under which overhead dominates.

10. Expected Challenges and Risk Mitigation

11. Work Plan and Timeline

Risk	Issue	Mitigation Strategy
Simplified aging model	Limited physical realism	Sensitivity analysis, alternative proxies, parameter sweeps
Synthetic workload bias	Incomplete representation of real behavior	Trace-driven workloads and multiple distribution families
RL instability	Reward sensitivity and training difficulty	Establish heuristic and supervised baselines first
Policy overhead	Control cost may offset benefits	Discrete action space and sparse update intervals
Fairness degradation	Aging protection may hurt some tasks	Fairness penalties and starvation bounds

Literature consolidation and proposal finalization

Baseline scheduler simulator

Thermal and aging model integration

Heuristic hybrid scheduler

Dataset generation and labeling

Supervised learning experiments

Bandit / RL experiments

Ablation, analysis, and final writing

12. Expected Outcomes

The expected outcome is that the proposed scheduler will demonstrate a better performance-reliability trade-off than conventional schedulers. The strongest experimental outcomes would be:

1. significant aging-score reduction under comparable throughput
2. lower waiting/turnaround time under the same aging limit
3. improved peak temperature and hotspot behavior
4. more consistent policy behavior across workload families

However, an objective evaluation must also report the following possibilities:

- gains may be small for some workloads

- some policies may reduce aging while worsening latency
- the best-performing policy and the most reliability-friendly policy may not be the same
- the RL-based version may not always outperform heuristic or supervised variants

Therefore, the purpose of the study is not merely to claim that the proposed method is universally best, but rather to show under which conditions it provides advantages, under which conditions it introduces cost, and how the trade-off surface is shaped.

13. Conclusion

This proposal presents a detailed and feasible research framework for bridging adaptive time quantum scheduling and aging-aware runtime control. The proposed system does not treat quantum selection purely as a performance problem; it explicitly incorporates hardware health into the decision process. In this sense, the work has the potential to make a meaningful contribution at the intersection of operating systems, computer architecture, and AI for systems.

References

- [1] H. S. Behera, R. K. Mohanty, and D. Nayak, “A New Proposed Dynamic Quantum with Re-Adjusted Round Robin Scheduling Algorithm and Its Performance Analysis,” *International Journal of Computer Applications*, 2010.
- [2] A. Noon, A. Kalakech, and S. Kadry, “A New Round Robin Based Scheduling Algorithm for Operating Systems: Dynamic Quantum Using the Mean Average,” arXiv preprint arXiv:1111.5348, 2011.
- [3] Sakshi, C. Sharma, S. Sharma, S. Kautish, S. A. M. Alsallami, E. M. Khalil, and A. W. Mohamed, “A New Median-Average Round Robin Scheduling Algorithm,” *Alexandria Engineering Journal*, 2022.
- [4] M. F. Zohora, F. Farhin, and M. S. Kaiser, “An Enhanced Round Robin Using Dynamic Time Quantum for Real-Time Asymmetric Burst Length Processes,” *PLOS ONE*, 2024.
- [5] P. Chitaliya et al., “Time Quantum Optimization in Round Robin Algorithm,” *NMIT-CON*, 2023.
- [6] J. Srinivasan et al., “The Case for Lifetime Reliability-Aware Microprocessors,” *ISCA*, 2004.
- [7] M. Agarwal et al., “Circuit Failure Prediction and Its Application to Transistor Aging,” *VTS*, 2007.
- [8] H. Aydin and D. Zhu, “Reliability-Aware Energy Management for Periodic Real-Time Tasks,” *IEEE Transactions on Computers*, 2009.
- [9] S. V. Kumar, C. H. Kim, and S. S. Sapatnekar, “Adaptive Techniques for Overcoming Performance Degradation Due to Aging in CMOS Circuits,” *IEEE Transactions on VLSI Systems*, 2011.
- [10] A. Yeganeh-Khaksar et al., “Ring-DVFS: Reliability-Aware Reinforcement Learning-Based DVFS for Real-Time Embedded Systems,” *IEEE Embedded Systems Letters*, 2021.